

ALEXANDER RAO

Software Engineer | Backend & Full-Stack Development | Machine Learning

PROFESSIONAL SUMMARY

Software engineer with experience building full-stack web applications, REST APIs, data-processing pipelines, and machine-learning systems. Strong foundation in Python, JavaScript, TypeScript, Java, SQL, React, Node.js, PostgreSQL, MongoDB, Docker, Git, and cloud deployment. Experienced in designing modular systems, integrating third-party APIs, testing production workflows, and translating business requirements into maintainable software.

TECHNICAL SKILLS

Languages	Python, Java, JavaScript, TypeScript, SQL, HTML, CSS
Frontend	React, Next.js, Redux, Tailwind CSS, Responsive UI
Backend	Node.js, Express.js, FastAPI, REST APIs, JWT, OAuth 2.0
Databases	PostgreSQL, MySQL, MongoDB, Redis, SQLAlchemy, Prisma
AI / ML	scikit-learn, pandas, NumPy, spaCy, Sentence Transformers, PyTorch, NLP
Tools	Git, GitHub, Docker, GitHub Actions, Postman, Linux, VS Code
Cloud	AWS EC2, S3, Lambda, CloudWatch, Render, Vercel
Engineering	OOP, DSA, system design, testing, CI/CD, Agile, code review

PROFESSIONAL EXPERIENCE

Software Engineer Intern — NovaStack Technologies | May 2025 – Aug 2025

- Developed and maintained Node.js and FastAPI services supporting customer onboarding and document-processing workflows.
- Designed REST endpoints, request validation, authentication middleware, structured error handling, and PostgreSQL data access layers.
- Reduced average API response time by approximately 28% by optimizing database queries, indexes, and repeated service calls.
- Implemented unit and integration tests with Pytest and Jest and participated in pull-request reviews and Git-based development.

Full-Stack Developer Intern — CodeHarbor Labs | Jan 2025 – Apr 2025

- Built React interfaces and Express.js APIs for an internal workflow management application used by multiple teams.
- Implemented JWT authentication, role-based access control, pagination, filtering, search, and reusable UI components.
- Integrated MongoDB with application services and created automated API tests using Postman collections and CI workflows.

PROJECTS

RecruitIQ — Requirement-Aware Intelligent Recruitment System | Python, NLP, FastAPI, React, PostgreSQL

- Designed an AI-assisted recruitment screening pipeline that extracts structured information from resumes and job descriptions.
- Implemented PDF text extraction, section detection, entity/skill extraction, semantic matching with Sentence Transformers, candidate ranking, and explainable scoring.
- Used PostgreSQL with structured relational data and JSON fields for flexible candidate metadata; exposed processing through REST APIs.

Real-Time Payment Fraud Detection Platform | Python, FastAPI, scikit-learn, PostgreSQL, React

- Built a transaction simulation environment for detecting suspicious payment behavior using historical and rolling-window features.
- Implemented transaction velocity, cumulative amount, recipient diversity, behavioral anomaly, and risk-scoring logic.
- Designed an investigation dashboard showing transaction context, risk factors, alerts, and detection latency.

Cloud-Native E-Commerce Platform | MERN, Redis, Docker, AWS

- Developed product, cart, order, payment, and authentication services using React, Node.js, Express, and MongoDB.
- Added Redis caching and asynchronous processing to reduce repeated database operations and improve application responsiveness.
- Containerized services with Docker and configured automated build and deployment workflows using GitHub Actions.

EDUCATION

Bachelor of Technology in Computer Science and Engineering — National Institute of Technology, India | 2023 – 2027 |
CGPA: 8.7/10

CERTIFICATIONS & COURSEWORK

AWS Certified Cloud Practitioner • Machine Learning Specialization • Data Structures and Algorithms • Database Management Systems • Computer Networks • Operating Systems • Software Engineering

ACHIEVEMENTS

- Finalist in a national-level software engineering hackathon focused on intelligent automation.
- Solved 350+ algorithmic problems across LeetCode and CodeChef.
- Contributed to open-source projects involving documentation, bug fixes, and API integrations.

ADDITIONAL INFORMATION

Languages: English, Hindi, Tamil **Interests:** distributed systems, applied NLP, developer tooling, cloud engineering